

# **PROJECT-1**

Factory-to-Customer Shipping  
Route Efficiency Analysis: A  
Case Study of Nassau Candy  
Distributor

# PROJECT REPORT

## Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

### Executive Summary

This report analyzes shipping performance across various routes, regions, and product lines for Nassau Candy Distributor. The goal is to identify inefficiencies, optimize delivery lead times, and improve logistics strategy.

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### Key Metrics Overview

- Total Orders: 9.689K
  - Total Lead Time: 13M days
  - Average Lead Time: 1.32K days
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### Total Shipping Lead Time by Ship Mode

- **Problem:** Standard Class shipping contributes the highest lead time (7.6M days), over 70% of total delay.
  - **Goal:** Reduce overall delivery time and improve customer satisfaction.
  - **Insight:** Faster modes like First Class (2.0M) and Same Day (0.7M) are underutilized despite offering quicker delivery.
  - **Recommendation:** Most shipments are sent through Standard Class, which takes the longest time. This slows down delivery and affects customer satisfaction. To improve, more orders should be sent using faster options like First Class and Same Day.
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### Total of Sales, Cost, and Gross Profit by Product Name

- **Problem:** A few products contribute most of the profit, while others like Wonka Gum (0.44%) barely generate returns.
- **Goal:** Focus on high-performing products to boost overall profit and reduce waste from low performers.
- **Insight:**
  - Top 3 products (Scrumdiddlyumptious, Fudge Mallows, Milk Chocolate) together contribute over 60% of total profit.

- Products like Wonka Gum, Kazookles, and Lickable Wallpaper show very low contribution.
  - **Recommendation:**
    - Some products like Scrumdiddlyumptious and Fudge Mallows make a lot of profit. Others like Wonka Gum make very little.
    - To earn more money and save costs, the company should focus on selling the popular products more.
    - Products that don't sell well should be checked — maybe they can be improved or stopped.
    - In short: **Sell more of what works, and fix or remove what doesn't.**
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## **Average Shipping Lead Time and Total Cost by Region**

- **Problem:** No region stands out as having a major delay or cost issue, which makes it harder to prioritize improvements.
  - **Goal:** Find regional differences in shipping performance to improve delivery speed or reduce cost.
  - **Insight:** All four regions (Atlantic, Gulf, Interior, Pacific) have nearly the same average lead time (~1.32K days) and cost (~25% each). There's no clear gap or outlier.
  - **Recommendation:** Since all regions perform similarly, focus on **route-level** or **factory-level** improvements instead of regional changes. Also, look at **product-level** or **shipping mode** differences to find better ways to reduce delays.
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## **Average Lead Time by Route**

- **Problem:** Some factory-to-region routes take much longer than others, causing delays in delivery.
- **Goal:** Find which routes are slow and improve them to speed up shipping.
- **Insight:**
  - The route **Sugar Shack → Pacific** has the longest average lead time (1,516.67 days).
  - Other routes are smaller in size, meaning they are faster or more efficient.
- **Recommendation:**

- Look into why Sugar Shack → Pacific is slow — maybe it's far, has transport issues, or low priority.
- Try sending Pacific orders from a different factory with faster delivery.
- Use this chart to find and fix other slow routes too.

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## **Additional information**

### **How many orders fall into each speed category:**

- Fast deliveries: 1.942K orders
- Moderate deliveries: 4.533K orders
- Slow deliveries: 3.214K orders

### **Which shipping modes are used for fast deliveries:**

- Standard Class: 1.10M lead time
- Second Class: 0.33M lead time
- First Class: 0.26M lead time
- Same Day: 0.08M lead time

👉 In simple words: Most fast deliveries are still happening through Standard Class, while First Class and Same Day (which are meant to be faster) are used much less.

### Which products are linked to slow deliveries?

Based on the donut chart filtered by slow delivery category, the following products are linked to slow deliveries and also generate high revenue:

- **Wonka Bar – Scrumdiddlyumptious:** 8.28K (22.16%)
- **Wonka Bar – Triple Dazzle Caramel:** 8.04K (20.21%)
- **Wonka Bar – Milk Chocolate:** 7.81K (19.63%)
- **Wonka Bar – Nutty Crunch Surprise:** 7.43K (18.68%)
- **Wonka Bar – Fudge Mallows:** 7.26K (18.25%)
- **Wonka Gum:** 0.20K (0.5%)

These top five products together contribute nearly 99% of the revenue in the slow delivery category.

### How to improve:

- Focus on speeding up delivery for the top five products listed above.
- Use faster shipping modes like **First Class** or **Same Day** for these products.
- Investigate which **routes or factories** are causing delays and reroute if needed.
- Products like **Wonka Gum** have low revenue and slow delivery — consider improving or replacing them.

# SOLVING BUSINESS QUESTIONS USING SQL-QUERIES

## Business Question

### **Which factory-to-customer routes are consistently efficient?**

The goal is to identify factory-location combinations that deliver orders quickly and reliably.

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## Query Logic

This SQL query analyzes shipping performance by joining sales data with factory metadata. Here's what it does:

- Joins distribution\_sales with factories using Product\_Name and Division.
  - Groups results by factory, city, and state/province.
  - Filters to only include routes with **5 or more orders** (to ensure consistency).
  - Calculates the **average shipping days** for each route.
  - Sorts the results in ascending order to highlight the fastest routes.
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## Recommendations

- Focus on improving slow-performing Lot's O' Nuts routes (e.g., Saginaw, Miramar, Westminster).
- Investigate what makes Hackensack and Midland efficient — replicate those practices.
- Consider rerouting orders from slow cities to faster ones if feasible.
- Use this analysis to prioritize logistics optimization by location, not just factory.

## Business Question

### Which factory-to-customer routes experience frequent delays?

The goal is to identify locations where shipping consistently takes longer than expected, so that logistics can be improved.

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#### Query Logic

This SQL query analyzes delayed shipping routes by:

- Joining `distribution_sales` with `factories` using `Product_Name` and `Division`.
  - Grouping results by factory, city, and state/province.
  - Filtering to include only routes with **5 or more orders** (to ensure reliability).
  - Calculating the **average shipping days** for each route.
  - Sorting results in **descending order** to highlight the slowest-performing routes first.
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#### Recommendations

- Prioritize investigation into **Lot's O' Nuts** routes, especially Saginaw, Miramar, and Westminster — they are the slowest.
- Check for issues like long distances, poor carrier performance, or factory bottlenecks.
- Consider rerouting orders from these cities through faster-performing locations.
- For **Wicked Choccy's**, monitor Burlington and Inglewood routes — they are slower than expected despite moderate order volumes.
- Use this data to optimize shipping schedules, improve customer satisfaction, and reduce operational lag.

## Business Question

### How does shipping performance vary by region, state, and shipping mode?

The goal is to understand which combinations of geography and shipping method lead to faster or slower deliveries.

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## Query Logic

This SQL query analyzes shipping performance across different regions and shipping modes:

- It groups the `distribution_sales` data by **Region, State/Province**, and **Ship\_Mode**.
  - Filters out combinations with fewer than 5 orders to ensure statistical reliability.
  - Calculates the **average shipping days** for each group.
  - Sorts the results in **descending order** to highlight the slowest-performing combinations first.
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## Recommendations

- Investigate why **Same Day** shipping is slow in Interior states like Wisconsin, Oklahoma, and Illinois.
- Review carrier performance and factory dispatch timing in **Louisiana** and **Rhode Island**, where First Class is underperforming.
- Consider switching to more reliable modes or optimizing warehouse-to-customer routing in these regions.
- Use this insight to prioritize logistics improvements by **region and shipping mode**, not just factory.



## Business Question

### Where do operational bottlenecks exist geographically?

The goal is to identify regions and states where shipping is both slow and expensive relative to sales — indicating inefficiencies.

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## Query Logic

This SQL query pinpoints geographic bottlenecks by:

- Grouping distribution\_sales by **Region** and **State/Province**.
  - Filtering for locations with **more than 5 orders** to ensure meaningful analysis.
  - Calculating:
    - **Average shipping days** — to measure delivery speed.
    - **Cost-to-sales ratio** — to measure operational efficiency.
  - Sorting results by **highest cost-to-sales ratio** and then by **longest shipping time**, to surface the worst-performing areas.
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## Recommendations

- Prioritize operational audits in **Nova Scotia, Manitoba, and Louisiana** — they show the worst cost-to-sales ratios and long delivery times.
- Investigate warehouse efficiency, carrier contracts, and route planning in these regions.
- Consider shifting fulfillment to nearby faster-performing states or optimizing inventory placement.
- Use this data to guide strategic decisions on logistics investment and cost control.

## Business Question

### Why does logistics optimization remain reactive instead of data-driven?

The goal is to uncover gaps in factory mapping that prevent proactive logistics planning.

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## Query Logic

This SQL query identifies products that are being shipped **without a mapped factory**, which signals a breakdown in data integration:

- Performs a **LEFT JOIN** between `distribution_sales` and `factories` using `Product_Name` and `Division`.
  - Filters for rows where `f.factory` IS NULL, meaning no factory is assigned.
  - Groups by product and division to count how many such unmapped orders exist.
  - Sorts by **total\_orders** to highlight the most problematic cases.
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## Recommendations

- Ensure every product in the sales system is properly mapped to a factory in the master data.
- Conduct a **data audit** to identify missing or mismatched entries between `distribution_sales` and `factories`.
- Automate alerts for unmapped products so logistics teams can fix issues before orders go out.
- Without this mapping, logistics decisions will remain reactive — based on complaints or delays — instead of being driven by performance data.